Polymer Processes

The most common type of polymer manufacturing is injection moulding, which is the focus of this topic. Other polymer manufacturing processes include extrusion, thermoforming, blow moulding and blown film processes.

Generally, polymer processes involve melting plastics to be injected into a mould, then closing it for the melt to cool, before being opened and ejected from the mould.

The raw plastic usually comes in two forms, pellets/resin and masterbatches. For the material, either thermoplastics, a low cost but high manufacturability and reversibility material, or thermosets, an irreversible thermal insulator with high strength and hardness, is used.

Parts of an Injection Molding Machine

- There are three major components:
- Injection/Plasticising Unit
  - feed plastic into the mould, with a hopper, feed throat, hydraulic cylinder, heater band and screw.
  - purpose is to melt and inject the plastic into the mold cavity
- Clamping Unit
  - clamps the mold together, with mechanical toggle or hydraulic methods.
  - keeps tightly closed under injection pressure
  - opens during ejection
- Control System Monitor and Control
  - controls:
    - position of plasticising unit, screw and mould
    - pressure during holding
    - velocity of screw during injection and mold
    - temperature of plasticising unit and mold.

Types of Injection Molding Machines

- the different types of injection molding machines are characterised by the type of clamping unit, being

- Toggle Clamp
  - fast, low cost, positive locking (actively locks????), however poor mold protection
- Hydraulic Clamp
  - good control and long term reliability with reduced size, and protects mould, but higher cost.

Phases of Injection Molding

- there are 5 stages to injection molding
- Plasticising
  - plastic pellets are fed into the injection screw and heated as they pass along the screw. heater bands and the shear forces by screw cause the plastic to melt, and is homogenised at the tip at the end of the plasticising unit.
  - As the screw is the main component moving and heating the plastic pellets, the type of screw used determines the melt quality.
  - Screws have three portions along its length, the feed, compression and metering section, where the thread depth generally decreases along its length
  - Conventional
    - cheap but has a poor melt quality
  - Barrier
    - deeper grooves than conventional screws, allowing increased melting rate and melt quality, but higher shear causes higher melt temp, thus more melt degradation
  - ET Screw
    - gives best melt quality for a wide range of polymers, but expensive
  - somethign something screw freezing experiment (wtf)
- 1st stage filling
  - mold filling involves filling the mold cavity as fast as possible by closing the mould and moving the injection unit forward to fill the mold volumetrically (???).
- 2nd stage packing

ensure pressure is effectively transmitted into the mould, to compensate for shrinkage and prevent backflow, until the gate has solidified.

cooling

molded part cools until sutable to be ejected, to prevent high temps from distorting the ejection pins.

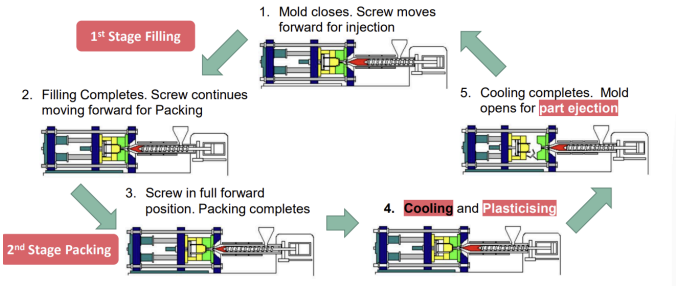
longest time of the cycle

screw rotates to plasticise the melt for the next cycle.

ejection

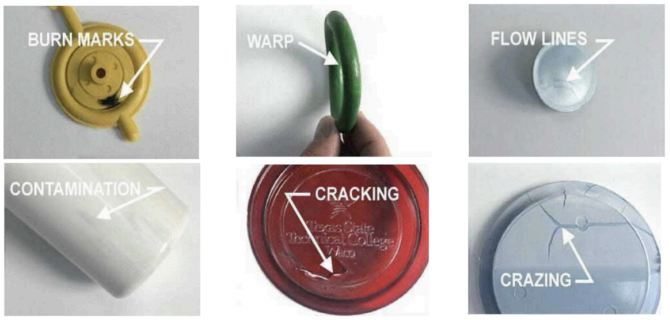
mold opens to eject part

The total time taken to fill, pack, cool and eject the part is known as the cycle time.



Molding Defects

Mold defects are similar to casting defects, due to the similarities in the process:



the main new ones are:

knit lines

jettings

burn marks

warps

contamination

crazing

Mold Design

The mainr purpose of an injection mold is to distribute melt, form it into the final shape, cool, and eject the part (similar to process of injection molding)

additionally, the mold must also be able to work mechanically with the other parts, meaning withstanding the molding forces, and transmit, guide and align its motion to the other moving parts.

Parts of a mold

Generally, there are three parts of a mold, the cavity (concave), the core (convex), and the parting line.

Either half may be the fixed half, or the moving half, the one connected to the clamping unit.

Some additional parts include the

Mold alignment - self locates the two halves together to prevent shifts

Feed system - molds the sprue, runner and gate.

Ejector system - has clearances for the ejector pins, return pins, support pillars, guide bushes and stop disc !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!1

Socket Head cap screws are typically used to join the assembly together (why is this here)

Types of Molds

Molds are typically categorised by a few categories:

- processed material
- basic design
- ejection method
- runner system
- number of cavities
- number of parting lines
- mold size

But the most common way to differentiate between moulds is the design type.

Two Plate Mold

- most common type of mold due to it being most simple, as there are only two mold halves (hence one parting line), only requiring opening in one direction, however limited to not have undercuts.

Stripper Plate Mold

- consists of three sandwiched plates, designed similar to a two plate mold but the middle plate is used for ejection, known as the stripper plate.
- typically for deep cup like shape moldings without undercut and with thin walls

Three Plate Mold

- a similar setup to stripper plate mold, with a fixed, floating and moving plate.
- because of the three plates, two parting lines are present, and the floating plates needs to be actuated seperately, giving a two step opening movement.
- Despite this, the molding and runner can often be automatically seperated, and the gate position is flexible, meaning it can be put at the bottom of the cup shaped molding

Sliding Side Core Mold

designed for parts with undercuts or external threads, the section with undercuts is made as a seperate sliding core, to allow the undercuts to not be damanged as it is slid out first, before the part is ejected with two plate mould.

sliders and cams are needed to actuate the lateral movement,

Hot Runner Mold

The mould has a heating element to heat the runner, for a higher quality part as there is no material loss from the runner system. the gate is also automatically seperated from the mold.

However, the heating element is much more expensive.